# Standard Operating Procedure for Resolving BGP Route Flap and OSPF Adjacency Loss on Cisco PE (MALL-PE-08)

**Node:** MALL-PE-08 | **Vendor:** Cisco | **Severity:** Critical | **Alarm Code:** ALARM_7197

## Description

This failure domain occurs on the PE role of Cisco equipment due to the inherent characteristics of Border Gateway Protocol (BGP) and Open Shortest Path First (OSPF) routing protocols. BGP route flapping is often caused by issues such as route oscillations, incorrect routing configurations, or hardware malfunctions. OSPF adjacency loss can be attributed to factors like network partitioning, authentication failures, or mismatched network topologies. ISIS LSP flooding, on the other hand, can be triggered by topology changes or routing misconfigurations. The Cisco PE device, MALL-PE-08, is equipped with specialized hardware and software that enables BGP, OSPF, and ISIS routing protocol support. However, when these protocols malfunction, it can lead to critical failure, requiring prompt resolution to maintain network uptime.

## Pre-requisite Checks

- Verify that all redundant links are operational and fully connected.
- Ensure that the Malleswaram Node site is accessible and physically secure.
- Confirm that the Cisco PE device, MALL-PE-08, is properly powered and online.
- Obtain necessary permissions and clearance for remote access, if required.

## Potential Root Causes

| Cause | Probability |
|---|---|
| BGP Route Flap due to incorrect routing configuration | High |
| OSPF Adjacency Loss due to network partitioning | Medium |
| ISIS LSP Flooding due to topology changes | Low |
| Hardware malfunction or equipment failure | Low |

## Diagnostic Commands & Expected Outputs

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 Routing table entries not generated Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PfxRcd 10.0.0.1 4 65000 104 103 1 0 0 00:01:10 ESTAB 4 , Description: This command displays a summary of BGP neighbor information, including the number of messages received and sent, as well as the table version.
```

```
Command: show ip ospf database, Expected Output: OSPF database summary LSA count: 10 Type 1 (Router): 5 Type 2 (Network): 3 Type 4 (Summary): 2 , Description: This command displays a summary of the OSPF database, including the number of LSA types and their counts.

Command: show isis database, Expected Output: ISIS database summary LSA count: 10 Type 10 (LSP): 5 Type 20 (SNP): 3 Type 30 (SNP): 2 , Description: This command displays a summary of the ISIS database, including the number of LSA types and their counts.
```

## Resolution Steps

Step 1: Step: Clear BGP session, Command: clear ip bgp *, Description: This command clears the BGP session, resetting the BGP table and sending a reset message to the neighbors.

Step 2: Step: Verify OSPF adjacency, Command: show ip ospf neighbor, Description: This command displays a list of OSPF neighbors, including their state and the number of LSA types they have received.

Step 3: Step: Reset ISIS database, Command: clear isis database, Description: This command resets the ISIS database, reinitializing the LSA counts and flushing the LSP cache.

## Post-Check Verification

```
Command: show ip bgp summary, Expected Output: BGP table version is 1, main routing table version 1 Routing table entries not generated Neighbor V AS MsgRcvd MsgSent TblVer InQ OutQ Up/Down State PfxRcd 10.0.0.1 4 65000 104 103 1 0 0 00:01:10 ESTAB 4 , Description: This command verifies that the BGP session has been successfully cleared and the BGP table has been reinitialized.

Command: show ip ospf database, Expected Output: OSPF database summary LSA count: 0 Type 1 (Router): 0 Type 2 (Network): 0 Type 4 (Summary): 0 , Description: This command verifies that the OSPF adjacency has been successfully re-established and the OSPF database has been flushed.

Command: show isis database, Expected Output: ISIS database summary LSA count: 0 Type 10 (LSP): 0 Type 20 (SNP): 0 Type 30 (SNP): 0 , Description: This command verifies that the ISIS database has been successfully reset and the LSA counts have been reinitialized.
```

## Roll-back Procedures

- Step: Re-establish BGP session, Command: clear ip bgp *, Description: This command re-establishes the BGP session, restoring the original BGP configuration and settings.

- Step: Re-initialize OSPF adjacency, Command: clear ip ospf neighbor, Description: This command re-initializes the OSPF adjacency, restoring the original OSPF configuration and settings.

- Step: Re-import ISIS database, Command: clear isis database, Description: This command re-imports the ISIS database, restoring the original ISIS configuration and settings.